

1 Solution — GIAM 3.1

Problem 13

1.1 Problem

Show that if two positive integers a and b satisfy $a \mid b$ and $b \mid a$, then they are equal.

1.2 Proof

***Proof.** Let a, b be positive integers with $a \mid b$ and $b \mid a$. By the definition of divides there are integers k, m with*

$$b = ak \text{ and } a = bm.$$

Substituting the first into the second,

$$a = bm = (ak)m = a(km).$$

Since $a > 0$ we have $a \neq 0$, so we may cancel a from both sides of $a \cdot 1 = a \cdot (km)$, giving

$$km = 1.$$

The only integers whose product is 1 are $k = m = 1$ and $k = m = -1$. If $k = -1$ then $b = -a < 0$, contradicting $b > 0$. Hence $k = 1$, and therefore $b = a \cdot 1 = a$. ■

1.3 Where Positivity Enters

Substituting each hypothesis into the other traps the two multipliers into a product of 1.

The proof

$$a \mid b \Rightarrow b = ak, k \in \mathbb{Z}$$

$$b \mid a \Rightarrow a = bm, m \in \mathbb{Z}$$

$$a = bm = (ak)m = a(km)$$

$$a > 0, \text{ so } a \neq 0 \text{ and we may cancel: } \mathbf{km = 1}$$

the only integers with $km = 1$ are $k = m = 1$ and $k = m = -1$

positivity rules out $k = -1$, so $k = 1$ and $b = a$ ■

Positivity is used twice

first: to cancel

$a > 0$ gives $a \neq 0$,
which licenses $a \cdot 1 = a \cdot (km) \Rightarrow km = 1$

second: to pick the sign

if $k = -1$ then $b = -a < 0$,
contradicting $b > 0$

Drop positivity and the theorem fails — you only get $a = \pm b$

$3 \mid -3$ and $-3 \mid 3$, yet $3 \neq -3$ (60 such pairs among nonzero $|x| \leq 30$)

This completes the partial order begun in Problem 11

property	statement	established in
reflexive	$a \mid a$	immediate ($k = 1$)
transitive	$a \mid b$ and $b \mid c \Rightarrow a \mid c$	Problem 11
antisymmetric	$a \mid b$ and $b \mid a \Rightarrow a = b$	this problem

so divisibility really is a partial order on the positive integers

Figure 1.1: Top panel, the proof: a divides b gives b equal to ak with k an integer; b divides a gives a equal to bm with m an integer; then a equals bm equals $(ak)m$ equals $a(km)$. Since a is positive it is nonzero and we may cancel, giving km equals 1. The only integers with km equal to 1 are k equals m equals 1 and k equals m equals minus 1. Positivity rules out k equals minus 1, so k equals 1 and b equals a . Two green panels: positivity is used twice — first to cancel, since a positive gives a nonzero which licenses cancelling a from a times 1 equals a times km ; second to pick the sign, since if k were minus 1 then b would equal minus a which is negative, contradicting b positive. A red panel: drop positivity and the theorem fails, giving only a equals plus or minus b — 3 divides minus 3 and minus 3 divides 3, yet 3 does not equal minus 3, with 60 such pairs among nonzero values up to 30 in absolute value. A final table completes the partial order begun in Problem 11: reflexive, a divides a , immediate with k equals 1; transitive, established in Problem 11; antisymmetric, highlighted, this problem. So divisibility really is a partial order on the positive integers.

The hypothesis “positive” is used at **two** distinct points, and it is worth separating them:

1. **To cancel.** From $a = a(km)$ we want $1 = km$, and cancellation requires $a \neq 0$. Positivity supplies that.
2. **To choose the sign.** Knowing $km = 1$ leaves two possibilities, $k = m = 1$ and $k = m = -1$. Positivity of b eliminates the second, since $k = -1$ would make $b =$

$-a$ negative.

Miss either one and the proof has a gap.

1.4 Remark — Without Positivity the Theorem Is False

Over all of $\mathbb{Z}$ the conclusion fails. Take $a = 3$, $b = -3$: then $3 \mid -3$ and $-3 \mid 3$, yet $3 \neq -3$. Exhaustive check over nonzero $|x| \leq 30$ finds 60 such pairs — and in every one, $a = -b$.

That is exactly what the proof predicts. Dropping positivity leaves $k = \pm 1$ unresolved, so the correct general statement over $\mathbb{Z}$ is

$$a \mid b \text{ and } b \mid a \implies a = \pm b, \quad \text{i.e.} \quad |a| = |b|.$$

The theorem as stated recovers the clean version by restricting to a set where the sign ambiguity cannot arise. This is a common pattern: a statement is *nearly* true in general, and the hypothesis is chosen precisely to remove the residual ambiguity.

1.5 Remark — An Alternative Proof by Size

A different route avoids the $km = 1$ analysis entirely, at the cost of one lemma.

Lemma. If $a \mid b$ with $a, b > 0$, then $a \leq b$.

Proof. Write $b = ak$. Since $a > 0$ and $b > 0$, the multiplier $k = b/a$ is positive; being a positive integer, $k \geq 1$. Hence $b = ak \geq a \cdot 1 = a$. $\square$

Now the theorem is immediate: $a \mid b$ gives $a \leq b$, and $b \mid a$ gives $b \leq a$, so $a = b$ by the antisymmetry of $\leq$. (Lemma verified on all positive pairs up to 200.)

This version is arguably more illuminating, because it explains **why** antisymmetry holds: divisibility among positives can only ever move *upward* in size, so a two-way divisibility forces the two numbers to sit at the same height. It also makes visible that divisibility-antisymmetry is *inherited* from $\leq$ -antisymmetry rather than being a new fact.

1.6 Remark — The Partial Order Is Now Complete

Problem 11 proved transitivity; this problem supplies antisymmetry; reflexivity is immediate ($a = a \cdot 1$). Together:

property	statement	source
reflexive	$a \mid a$	immediate, $k = 1$
transitive	$a \mid b$ and $b \mid c \Rightarrow a \mid c$	Problem 11
antisymmetric	$a \mid b$ and $b \mid a \Rightarrow a = b$	this problem

Table 1.1

So divisibility is genuinely a **partial order** on $\mathbb{Z}^+$ — the fact asserted back in Problem 1.7.1 and now fully proved. It is *partial* rather than total because incomparable pairs exist: neither $4 \mid 6$ nor $6 \mid 4$. Contrast $\leq$, under which any two integers are comparable.

1.7 Remark — Why the Only Factorisations of 1 Are ± 1

The step “the only integers with $km = 1$ are ± 1 ” deserves a word, since the whole proof turns on it. If $|k| \geq 2$ then $|km| = |k||m| \geq 2|m|$, which is either 0 (if $m = 0$) or at least 2 — never 1. The same holds with roles swapped, so both $|k|$ and $|m|$ equal 1, and their product being positive forces matching signs.

In the language of ring theory, this says 1 and -1 are the only **units** of $\mathbb{Z}$. That fact is what makes divisibility in $\mathbb{Z}$ well behaved: it is precisely why “ $a \mid b$ and $b \mid a$ ” pins the pair down to a sign, and why unique factorisation is stated only up to units. In rings with more units the ambiguity is correspondingly larger — in the Gaussian integers, where the units are $\pm 1, \pm i$, mutual divisibility gives $a = \pm b$ or $a = \pm ib$.

1.8 Remark — Cancellation Is Not Free

The step from $a \cdot 1 = a \cdot (km)$ to $1 = km$ is cancellation, and it needs $a \neq 0$ — otherwise both sides read $0 = 0$ and nothing follows. This is the same caution as multiplying an inequality (Problem 6) or clearing a denominator (Problem 12): an operation that looks purely mechanical carries a side condition.

Cancellation is valid in $\mathbb{Z}$ for any nonzero factor precisely because $\mathbb{Z}$ has **no zero divisors** — the property already needed in Problem 3. In a ring where $2 \cdot 3 = 0$ is possible, such as $\mathbb{Z}_6$, cancelling is illegitimate and this proof would collapse.